

$$\dot{P} = (D_T \nabla^2 + D_R \partial_\phi^2 - V_A \hat{u}(\phi) \cdot \nabla) P(\vec{r}, \phi) - \gamma^{-1} \nabla \cdot (P(\vec{r}, \phi) \vec{I}[P](\vec{r}, \phi))$$

$$\vec{I} = \int r dr d\psi_R d\phi' U'(r) \hat{u}(\psi_R) g(r, \theta_1, \theta_2) P(\vec{r} + r \hat{u}(\psi_R), \phi')$$

$$\begin{aligned} P(\vec{r} + r \hat{u}(\psi_R)) &= P(\vec{r} + r \cos \theta_1 \hat{u} + r \sin \theta_1 \hat{v}) \\ &= \sum_n \frac{1}{n! m!} (\hat{u} \cdot \nabla)^n (\hat{v} \cdot \nabla)^m P(\vec{r}) r^{n+m} \cos^n \theta_1 \sin^m \theta_1 \\ &\equiv D_{nm} P(\vec{r}) \end{aligned}$$

$$\begin{aligned} &= \sum_{\substack{n, m \\ n_1, n_2}} \frac{1}{n! m!} \int dr d\psi_R d\phi' U'(r) (\cos \theta_1 \hat{u} + \sin \theta_1 \hat{v}) D_{nm} P(\vec{r}, \phi') r^{n+m+1} \cos^n \theta_1 \sin^m \theta_1 \\ &\quad g_{n, n_2}(r) \cos(n_1 \theta_1 + n_2 \theta_2) \end{aligned}$$

$$P(\phi') = \sum_{\ell'} P_{\ell'} e^{i \ell' \phi'}, \quad P_{\ell'} = P_{-\ell'}$$

$$\begin{aligned} &= \sum_{\ell'} D_{nm} \int r^{n+m+1} U'(r) g_{n, n_2}(r) \cdot \cos^n \theta_1 \sin^m \theta_1 (\cos \theta_1 \hat{u} + \sin \theta_1 \hat{v}) \\ &\quad \cos(n_1 \theta_1 + n_2 \theta_2) P_{\ell'}(\vec{r}) e^{i \ell' (\theta_2 + \phi)} \end{aligned}$$

$$I_{\theta_2} = \int_0^{2\pi} \cos(n_1 \theta_1 + n_2 \theta_2) e^{i \ell' \theta_2} d\theta_2$$

$$= \frac{e^{i n_1 \theta_1}}{2} \int_0^{2\pi} e^{i (\ell' + n_2) \theta_2} d\theta_2 + \frac{e^{-i n_1 \theta_1}}{2} \int_0^{2\pi} e^{i (\ell' - n_2) \theta_2} d\theta_2$$

$$= \pi e^{i n_1 \theta_1} \delta_{\ell', -n_2} + \pi e^{-i n_1 \theta_1} \delta_{\ell', n_2} = \pi e^{\pm i n_1 \theta_1} \delta_{\ell', \mp n_2}$$

$$I_{\theta_1} = \int_0^{2\pi} \cos^{n+1} \theta_1 \sin^{m+1} \theta_1 \cdot \pi e^{\pm i n \theta_1} d\theta_1 \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix}$$

$$\equiv \underline{\underline{C}}(n, m, \pm n_1) \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} = \begin{pmatrix} \hat{u} C(n+1, m, \pm n_1) \\ \hat{v} C(n, m+1, \pm n_1) \end{pmatrix}$$

$$I_r = \int_0^\infty V'(r) g_{n,n_2}(r) r^{n+m+1} dr \equiv A(n, m, n_1, n_2)$$

$$= \sum_{n, m, n_1, n_2} A(n, m, n_1, n_2) \underline{\underline{C}}(n, m, \pm n_1) e^{\mp i n_2 \phi} \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} D_{nm} \rho_{\mp n_2}(\vec{r})$$

$$I_{\hat{u}} = \sum_{\substack{n, m, n_1, n_2 \\ -\infty \sim \infty}} A(n, m, n_1, n_2) C(n+1, m, \pm n_1) e^{\mp i n_2 \phi} D_{nm} \rho_{\mp n_2}(\vec{r})$$

$$= \sum_{n, m, n_1, n_2} A(n, m, n_1, \mp n_2) C(n+1, m, \pm n_1) e^{\pm i n_2 \phi} D_{nm} \rho_{n_2}(\vec{r})$$

$$A(n, m, n_1, -n_2) = \int_0^\infty V'(r) \underline{g_{n_1, -n_2}}(r) r^{n+m+1} dr$$

$$g(r, \theta_1, \theta_2) = \sum_{n_1, n_2} g_{n_1, n_2}(r) \cos(n_1 \theta_1 + n_2 \theta_2)$$

$$\cos(n_1 \theta_1 \pm n_2 \theta_2) = \underbrace{\cos n_1 \theta_1 \cos n_2 \theta_2}_{\text{sym}} \mp \underbrace{\sin n_1 \theta_1 \sin n_2 \theta_2}_{\text{anti}}$$

$$g = \sum_{\substack{n_1, n_2 \\ n_1 > 0, n_2 \in \mathbb{Z}}} g_{n_1, n_2} (\cos n_1 \theta_1 \cos n_2 \theta_2 \mp \sin n_1 \theta_1 \sin n_2 \theta_2)$$

$$\begin{aligned}
&= \sum_{n_1, n_2} g_{n_1, n_2} (\cos n_1 \theta_1 \cos n_2 \theta_2 \mp \sin n_1 \theta_1 \sin n_2 \theta_2) \\
&\quad + g_{n_1, -n_2} (\cos n_1 \theta_1 \cos n_2 \theta_2 \pm \sin n_1 \theta_1 \sin n_2 \theta_2) \\
&\quad n_1, n_2 > 0 \\
&= \sum (g_{n_1, n_2} + g_{n_1, -n_2}) \cos - (g_{n_1, n_2} - g_{n_1, -n_2}) \sin \quad n_1, n_2 > 0 \\
&\equiv B_s(n_1, n_2) \cos n_1 \theta_1 \cos n_2 \theta_2 \quad \left| \quad B_s = g_{n_1, n_2} + g_{n_1, -n_2} \right. \\
&\quad + B_a(n_1, n_2) \sin n_1 \theta_1 \sin n_2 \theta_2 \quad \left| \quad B_a = g_{n_1, n_2} - g_{n_1, -n_2} \right. \\
&g_{n_1 \pm n_2} = \frac{1}{2} (B_s \mp B_a) \Rightarrow A(\pm n_2) = \frac{1}{2} (A_s \mp A_a)
\end{aligned}$$

$$\begin{aligned}
&= \sum_{n, m, n_1, n_2} \underline{C(n+1, m, \pm n_1) (A_s \overset{\pm |n_2|}{\pm} A_a)} D_{nm} e^{i n_2 \phi} \rho_{n_2} \\
&\quad = \frac{1}{2} [C(A_s \pm A_a) + C^*(A_s \mp A_a)] \quad n_2 > 0 \\
&\quad = \frac{1}{2} [A_s \underset{2C_R}{(C+C^*)} \pm A_a \underset{2iC_I}{(C-C^*)}] \quad n_2 < 0
\end{aligned}$$

$$= \sum_{n, m, n_1, n_2} D_{nm} (C_R A_s + i C_I A_a) e^{i n_2 \phi} \rho_{n_2}$$

$$= \sum_{n, m, n_1, n_2} D_{nm} (C_R A_s(n_2) + i C_I A_a(n_2)) e^{i n_2 \phi} \rho_{n_2}$$

$$\begin{aligned}
&n_2 > 0 \quad + D_{nm} (C_R A_s(+n_2) + i C_I A_a(+n_2)) e^{i n_2 \phi} \rho_{n_2} \\
&\quad + D_{nm} (C_R A_s(-n_2) - i C_I A_a(-n_2)) e^{-i n_2 \phi} \rho_{-n_2}
\end{aligned}$$

$$= \sum_{nmn_1n_2} D_{nm} \left[C_R A_s (e^{in_2\phi} P_{n_2} + e^{-in_2\phi} P_{-n_2}) \right. \\ \left. + i C_I A_a (e^{in_2\phi} P_{n_2} - e^{-in_2\phi} P_{-n_2}) \right]$$

$$= \sum_{nmn_1n_2} D_{nm} \left[C_R A_s (P_{n_2} \cos n_2\phi + Q_{n_2} \sin n_2\phi) \right. \\ \left. + C_I A_a (Q_{n_2} \cos n_2\phi - P_{n_2} \sin n_2\phi) \right]$$

$$P \cos \chi + Q \sin \chi = \sqrt{P^2 + Q^2} \sin \left(\chi + \tan^{-1} \frac{P}{Q} \right)$$

$$Q \cos \chi - P \sin \chi = \sqrt{P^2 + Q^2} \cos \left(\chi + \tan^{-1} \frac{P}{Q} \right)$$

χ φ

$$= \sum_{nmn_1n_2} D_{nm} X_{n_2} \left(\underline{C_R A_s} \sin(n_2\phi + \varphi_{n_2}) \right. \\ \left. + \underline{C_I A_a} \cos(n_2\phi + \varphi_{n_2}) \right)$$

$$= \sum_{nmn_1n_2} D_{nm} X_{n_2} \underline{Y_{C,A}} \sin(n_2\phi + \underbrace{\varphi_{n_2}}_{P_{n_2}(\vec{r})} + \underbrace{\eta_{C,A}}_{\substack{n.m.n_1.n_2.[g] \\ \text{const.}}})$$

General Form

passive: $P(\vec{r}, \phi) = P_0(\vec{r})$

$g(r, \theta_1, \theta_2) = g_{00}(r)$

$$A(n, m, n_1, n_2) = 0 \Rightarrow A_{s/a}(n_1, n_2) = 0$$

$$Y(n, m, n_1, n_2) = \sqrt{C_R^2 A_s^2 + C_I^2 A_a^2} = 0$$

$$\tan \eta_{c,A} = \frac{C_I A_a}{C_R A_s} = \text{Arg}(C(n+1, m, n_1)) \cdot \frac{A_a(n, m, n_1, n_2)}{A_s(n, m, n_1, n_2)}$$

Phase shift of $\cos^{n+1} \theta \sin^m \theta$?
on n_1 mode

$$Y(n, m, n_1, n_2) = \sqrt{C_R^2 A_s^2 + C_I^2 A_a^2}$$

$$\rho(\vec{r}, \phi) = \sum_{n_2} X_{n_2}(\vec{r}) \sin(n_2 \phi + \varphi_{n_2}(\vec{r}))$$

$$\Rightarrow I \hat{u} = \sum_{nmn_1} D_{nm} H(Y_{c,A}, \eta_{c,A}) \rho$$

$\uparrow \quad \uparrow$
 filtering dispersion

$$C(a, b, r) = \frac{2\pi}{2^{a+b} i^b} \sum_S \binom{a}{p} \binom{b}{q} (-1)^q$$

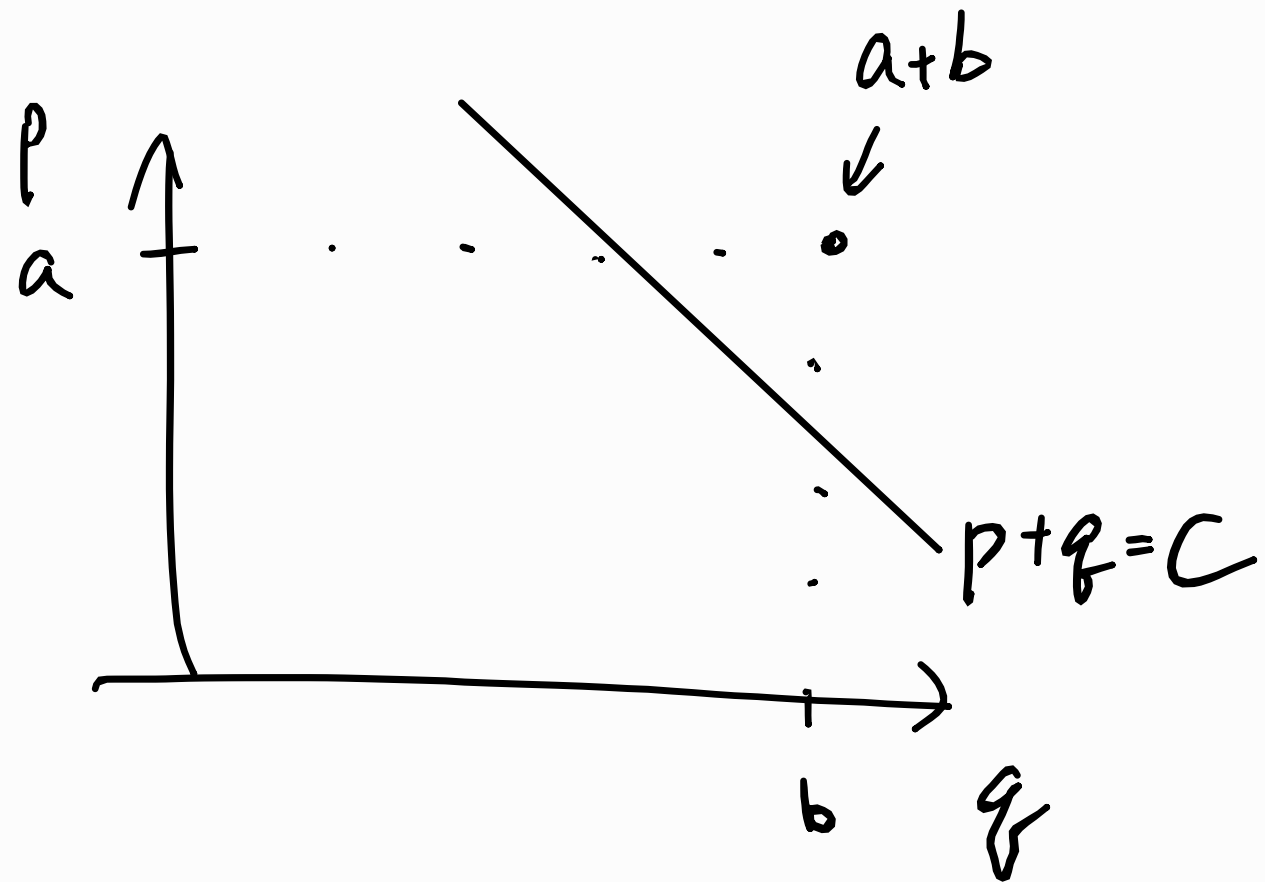
$$S: 0 \leq p \leq a, 0 \leq q \leq b, p+q = \frac{1}{2}(r+a+b)$$

$$\Rightarrow \frac{1}{2}(r+a+b) \in [0, a+b]$$

$$\Rightarrow 0 \leq \frac{1}{2}(r+a+b) \leq a+b$$

$$\Rightarrow -(a+b) \leq r \leq a+b$$

$$C(a, b, -r) = C^*(a, b, r)$$



$$C(n, m, n, \pm 1) = \int_0^{2\pi} \cos^n \theta \sin^m \theta e^{in\theta} e^{\pm i\theta} d\theta$$

$$= \int_0^{2\pi} \cos^n \theta \sin^m \theta e^{in\theta} (\cos \theta \pm i \sin \theta) d\theta$$

$$= C(n+1, m, n, \pm 1) \pm i C(n, m+1, n, \pm 1)$$

$$= (C_R(n+1, m) + i C_I(n+1, m))$$

$$\pm i (C_R(n, m+1) + i C_I(n, m+1))$$

$$C_R(n, m, n, \pm 1) = C_R(n+1, m, n,) \mp C_I(n, m+1, n,)$$

$$C_I(n, m, n, \pm 1) = C_I(n+1, m, n,) \pm C_R(n, m+1, n,)$$

$$C_R^\pm \equiv C_R(n, \pm 1)$$

$$C_R^+ + C_R^- = 2C_R(n+1) \quad C_I^+ + C_I^- = 2C_I(n+1)$$

$$C_I^+ - C_I^- = 2C_R(m+1) \quad C_R^- - C_R^+ = 2C_I(m+1)$$

$$C_R(n+1) \pm C_R(m+1) = \frac{1}{2} (C_R^+ \pm C_I^+ + C_R^- \mp C_I^-)$$

$$C_I(n+1) \pm C_I(m+1) = \frac{1}{2} (C_I^+ \mp C_R^+ + C_I^- \pm C_R^-)$$

$$\underline{\hat{I}} = \sum_{nm, n_2} D_{nm} X_{n_2} \left[\hat{u} \left(A_s C_R(n+1, m) \sin(n_2 \phi + \varphi_{n_2}) + A_a C_I(n+1, m) \cos(n_2 \phi + \varphi_{n_2}) \right) + \hat{v} \left(A_s C_R(n, m+1) \sin(n_2 \phi + \varphi_{n_2}) + A_a C_I(n, m+1) \cos(n_2 \phi + \varphi_{n_2}) \right) \right]$$

$$\hat{W}_+ \equiv \frac{\hat{u} + \hat{v}}{2}$$

$$\hat{W}_- \equiv \frac{\hat{u} - \hat{v}}{2}$$

$$= \sum_{nm, n_2} D_{nm} X_{n_2} \left[A_s \sin(n_2 \phi + \varphi_{n_2}) (C_R(n+1, m) \hat{u} + C_R(n, m+1) \hat{v}) + A_a \cos(n_2 \phi + \varphi_{n_2}) (C_I(n+1, m) \hat{u} + C_I(n, m+1) \hat{v}) \right]$$

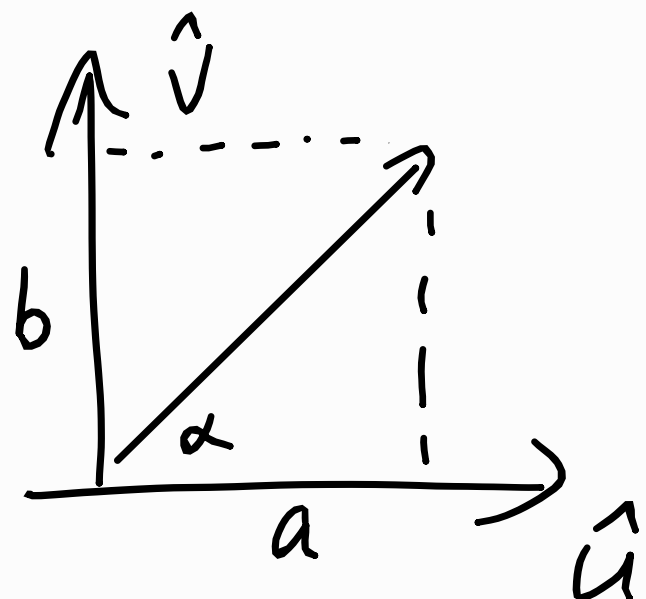
$$= \frac{1}{2} \sum_{nm, n_2} D_{nm} X_{n_2} \left[A_s \sin(n_2 \phi + \varphi_{n_2}) (C_R^+ \pm C_I^+ + C_R^- \mp C_I^-) \hat{W}_\pm + A_a \cos(n_2 \phi + \varphi_{n_2}) (C_I^+ \mp C_R^+ + C_I^- \pm C_R^-) \hat{W}_\pm \right]$$

$$\hat{W}_+ \left((R^+ + I^+ + R^- - I^-)s + (I^+ - R^+ + I^- + R^-)c \right)$$

$$\hat{W}_- \left((R^+ - I^+ + R^- + I^-)s + (I^+ + R^+ + I^- - R^-)c \right)$$

$$\Rightarrow \hat{u} \left((R^+ + R^-)s + (I^+ + I^-)c \right)$$

$$\hat{v} \left((I^+ - I^-)s - (R^+ - R^-)c \right)$$



$$\hat{x} = \cos \theta \hat{u} + \sin \theta \hat{v}$$

$$a \cos \theta \hat{u} + b \sin \theta \hat{v}$$

$$= a \hat{x} + (b-a) \sin \theta \hat{v}$$

$$= b \hat{x} + (a-b) \cos \theta \hat{u}$$

